“PAPER_{0:D3}” -f [int]# PAPER #120 — UQFF Astronomical Systems Catalog: 24-System Parameter Reference

Title: Complete Catalog of Astronomical Systems Used in UQFF Calculations — Parameters, Verification Sources, and Equation Assignments

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Framework: UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57, ß_i = 0.61)
Date: March 2026
Source: Grok thread `2fe4fa3e` — DeepSearch extraction of “UQFF Equations Across Astrophysical Systems_22Sept2025.docx” (393 pages)
Index Slot: §1.16 UQFF Equation Systems Reference

$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4}\,\text{day}^{-1},\ [SSq] = 0.57,\ \beta_i = 0.61$$

Abstract

The UQFF framework has been validated and applied across 24 unique astrophysical systems spanning stellar, galactic, nuclear, particle, and cosmological domains. This paper provides a complete parameter catalog for all 24 systems, organized by type, with UQFF equation assignments, calibrated parameter values, verification data sources, and cross-references to existing whitepapers. The catalog is extracted from the 393-page “UQFF Equations Across Astrophysical Systems” corpus (verified Sept 22, 2025) as the authoritative single-document reference for system parameters used in any UQFF calculation.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10?4 day?¹, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. Stellar and Solar Systems

1.1 Sun (Solar System)

Role in UQFF: Heliosphere modeling via Ug2 bubble, solar wind calibration.

Parameter	Value	Source
M_s	1.989×10^{30} kg	IAU 2015
R_b (heliosphere bubble radius)	1.496×10^{13} m = 100 AU	Voyager 1 crossing
ω_s (rotation rate)	2.5×10^{-6} rad/s	Solar sidereal rotation
B_s (surface magnetic field)	0.0001–0.4 T	GOES 2025
δ_{sw}	$0.01 = [SSq]/57$	EP-07, PSP CDAWeb E01–E17
v_{sw}	5×10^5 m/s	EP-07 PSP baseline
ρ_{sw}	$\approx 8 \times 10^{-21}$ kg/m³	Parker Solar Probe
H_{SCm}	≈ 1	Heliosphere SCm factor

Parameter	Value	Source
T_s (surface temperature)	5778 K	IAU

UQFF equations: U_{g2} (bubble), U_m (disk strings), $U_{b,i}$ (expansion)
Verification: EP-07 (PAPER_114), Parker Solar Probe CDAWeb 2025
Q_wave contribution: $Q_{wave,\odot}$ included in 47-system mean 3.97×10^4 J/m³

1.2 Westerlund 2 (Star Cluster)

Role in UQFF: Triadic master for turbulent outflow modeling.

Parameter	Value	Source
Distance	≈ 8 kpc	Gaia DR3
Mass	$\sim 10^4$ M?	JWST 2025
Age	~ 2 Myr	Membership surveys
SFR	High (active star formation)	JWST 2024
Q_{wave}	$\approx 3.97 \times 10^4$ J/m³ (mean stack)	47-system array

UQFF equations: $F_{U,\text{Triadic}}$ with $n = 13$ (plasma), U_m (turbulence)
Verification: JWST 2025 images, 47-system Q_wave stats (Jarque-Bera = 8.78, $p = 0.012$)
Cross-reference: Q_wave_47 statistical system

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2. Black Hole and Galactic Center Systems

2.1 Sagittarius A* (Sgr A*)

Role in UQFF: SMBH anchor for U_{g4} (BH-star interaction), galactic distance d_g.

Parameter	Value	Uncertainty	Source
M_{bh}	8.15×10^{36} kg = 4.1×10^6 M?	±4.68%	Gaia DR4 2025
d_g	2.44×10^{20} m = 25,800 ly	±4.51%	Gaia DR4 (PAPER_110)
d_g (UQFF)	2.55×10^{20} m = 27,000 ly	—	UQFF calibrated
ω_g	7.3×10^{-16} rad/s	—	Galactic rotation curve
U_{g4}	1.8937×10^{-23} J/m³	—	PAPER_110 calculated

UQFF equations: U_{g4}, E_{react} (from BH-scale reactivity)
Verification: EP-06 (PAPER_110), Gaia DR3/DR4 2025
Note: The UQFF uses $d_g = 2.55 \times 10^{20}$ m (Galactic longitude model) vs Gaia's 2.44×10^{20} m (4.5% difference within observational uncertainty).

2.2 G359.13142-0.20005

Role in UQFF: High-z object for shear map analysis and triadic analogs.

Parameter	Value	Source
Galactic coordinates	$l = 359.13°$, $b = -0.20°$	NISP survey
δ_τ	≈ 0.05	JWST NISP 2025

UQFF equations: $F_{U,\text{Triadic}}$, shear maps, δ_τ calibration
Verification: JWST 2025 data, NISP catalog

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3. Quasar and Blazar Systems

3.1 3C 273 (Quasar)

Role in UQFF: Asymmetric jet verification of t_n reversals.

Parameter	Value	Source
Redshift z	0.158	Schmidt 1963
Luminosity L	$\sim 10^{46}$ erg/s = 10^{39} W	MNRAS 2025
Jet length	~ 200 kpc (apparent), 65 kpc proper	VLBI
Brightness asymmetry	$> 100 : 1$	MNRAS 482, 743 (2019)
N (reversal count)	13 sequential t_n reversals	PAPER_115
Asymmetry ratio R	$130 > 100$ target	EP-09 calculation

UQFF equations: $U_{b,i}$ with $\cos(\pi t_n)$, U_m; ratio $R = |\cos(\pi t_{n1})/\cos(\pi t_{n2})|^N$
Verification: EP-09 (PAPER_115), MNRAS 482/743 2019, arXiv:2412.18250

3.2 RACS J0320-35 (Quasar)

Role in UQFF: Navier-Stokes jet fluid verification of $U_{b,i}$ asymmetry.

Parameter	Value	Source
Jet asymmetry ratio R	≈ 1.5	Chandra 2025
τ_{dissip}	9 Gyr	EP-01 calculation
$\cos(\omega t_n)$ reversal	Sign flip confirmed	EP-01

UQFF equations: $U_{b,i}$, Navier-Stokes fluid in jets
Verification: EP-01 (PAPER_111), Chandra X-ray Observatory 2025

3.3 Blazars (Fermi LAT 4LAC)

Role in UQFF: E_{react} decay rate κ calibration across 3,743 sources.

Parameter	Value	Source
Sample size	3,743 sources (4LAC-DR3)	Fermi HEASARC
Luminosity range	10^{39}–10^{47} W	4LAC-DR3
κ (measured)	$0.000497 \pm 5\%$ day?¹	EP-05 PAPER_113
8-bin κ error	All $< 5\%$	EP-05
E_{react} range	10^{39}–10^{47} W/m³ (8 L bins)	EP-05

UQFF equations: $E_{react} = 10^{46}e^{-\kappa t}$, $L \propto E_{react}$ by blazar class
Verification: EP-05 (PAPER_113), Fermi LAT 4LAC-DR4, HEASARC 2025

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4. Galaxy Cluster and Radio Systems

4.1 PLCK G287.0+32.9 (Galaxy Cluster)

Role in UQFF: Double radio relic calibration for triadic systems.

Parameter	Value	Source
Type	Galaxy cluster (double relic)	Planck PSZ2
$M_{500,X}$	Calibrated to triadic mass scale	PSZ2
Q_wave	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$, $M_{500,X}$ calibration
Verification: Planck/PSZ2 catalog

4.2 ASKAP J1832-0911 (Radio Transient)

Role in UQFF: Transient source for Q_wave array updates and triadic analogs.

Parameter	Value	Source
Survey	ASKAP 2025	ASKAP
Classification	Radio transient	ASKAP
Q_wave	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$ transient variant
Verification: ASKAP 2025 survey

4.3 PSZ2 G181.06+48.47 (Galaxy Cluster)

Role in UQFF: Low-mass merger for double relic analogs.

Parameter	Value	Source
$M_{500,X}$	2.57×10^{14} M?	Planck 2025
Merger type	Low-mass merger	Planck
Q_wave	In 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$, merger relic analog
Verification: Planck 2025, PSZ2 catalog

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5. Transient and Merger Systems

5.1 GW170817 (BNS Merger)

Role in UQFF: β_i calibration via r-process ejecta fraction.

Parameter	Value	Source
Event date	August 17, 2017	LIGO/Virgo
Total merger mass	$\approx$ 2.8 M?	LIGO 2017
Ejecta mass M_{ej}	$\approx$ 0.05 M? total	Spectral fit
Dynamical fraction	40% ($\approx 1 - \beta_i$)	EP-11
Ejecta velocity	$0.1c$ (blue), $0.3c$ (red)	Kilonova spectral
Y_e	$\approx$ 0.1 (neutron-rich)	EP-11 calibration
r-process fraction A > 140	$\approx 95\%$ solar	EP-11
$U_{b,i}$ (calculated)	$\approx 2.3 \times 10^{-3}$ M?/Gyr	EP-11

UQFF equations: $U_{b,i}$ feeding outflows, $Y_e = 1/(1 + e^{[SCm]/[UA]})$
Verification: EP-11 (PAPER_109), LIGO/Virgo 2017 + 2025 NR simulations

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5.2 AT2024tvd (Astrophysical Transient)

Role in UQFF: NS merger analog for triadic system update (2024).

Parameter	Value	Source
Discovery	2024	ATel/ZTF
Classification	NS merger analog	arXiv 2506.04440
Q_wave update	Extends 47-system array	Grok corpus

UQFF equations: $F_{U,\text{Triadic}}$ transient variant
Verification: arXiv:2506.04440, ZTF 2024

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6. Exoplanet Systems

6.1 TOI 1227 b (Exoplanet)

Role in UQFF: $U_{b,i}$ calibration for planetary mass loss.

Parameter	Value	Source
Mass loss rate	$\approx 10^{12}$ g/s	TESS/JWST 2025
System	Young transiting exoplanet	TESS
Q_wave	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ (atmospheric escape), $F_{U,\text{Triadic}}$ for mass loss dynamics
Verification: TESS/JWST 2025

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7. Nebula and Star Formation Systems

7.1 Pillars of Creation (Nebula)

Role in UQFF: Buoyancy refinement for star-forming outflows in Ug2/Ub_i.

Parameter	Value	Source
Location	M16 (Eagle Nebula)	—
Distance	$\approx 6,500$ ly	Gaia
Type	Active star formation, pillar instability	JWST 2022/2024
Q_wave	In 47-system triadic array	Grok corpus

UQFF equations: $U_{b,i}$ buoyancy refinements, U_{g2} bubble, alpha-conjugate clustering
Verification: JWST 2025, EP-12 alpha BEC (PAPER_107)

7.2 Nebular Dynamics (Generic)

Role in UQFF: Dust/gas stability modeling via U_i.

Parameter	Value	Source
Context	UQFF Drawing 32 Nebular system	Internal
Stability variable	$U_i = \lambda_i \rho_{vac,[SCm]} \rho_{vac,[UA]} \omega_s \cos(\pi t_n)$	Framework

UQFF equations: U_i (universal inertia for dust/gas stability)

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8. Nuclear and Particle Systems

8.1 Pb-206 (Nuclear System)

Role in UQFF: Nuclear binding energy ladder, $[SSq]$ calibration.

Parameter	Value	Source
n (level)	8.2 ($\Delta n = 0.21$)	EP-04
S_n (neutron separation)	$\approx 2 \times [SSq] \times E_8$ (3.5% error)	EP-04

Parameter	Value	Source
Level spacing	~ 10 MeV = 10^{-12} J	ENSDF/NNDC 2025
Polynomial fit degree	$n = 8$ ($R^2 \approx 0.95$)	EP-04
Shell levels	10+ levels in NNDC database	ENSDF

UQFF equations: $E_n = E_0 \times 10^n$ ($n = 8$), $[SSq] = 0.57$ calibration point
Verification: EP-04 (PAPER_117), ENSDF/NNDC 2025

8.2 ¹²C Hoyle State (Nuclear BEC)

Role in UQFF: Bose term N_B calibration in U_m.

Parameter	Value	Source
Energy	7.65 MeV	Hoyle 1954
Configuration	3-alpha BEC ($N_B = 3$)	Tohsaki AMD
Condensate fraction	70–90%	Tohsaki et al. 2001
T_c (nuclear BEC)	$\approx 1.2 \times 10^6$ K	EP-12 calculation
LENR ΔT_c shift	≈ 300 K	EP-12 UQFF
¹6O analog	4-alpha BEC ($N_B = 4$)	Funaki et al.

UQFF equations: N_B term in U_m, LENR T_c shift via E_{react} scaling
Verification: EP-12 (PAPER_107), Tohsaki et al. arXiv:1103.3940

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9. Compact Object Systems

9.1 Magnetar SGR 1745-2900

Role in UQFF: Full g_{Magnetar} equation anchor; B/B_{crit} calibration.

Parameter	Value	Source
Type	Soft gamma repeater magnetar	Chandra/Swift
B field	$\sim 10^{10}$–10^{11} T	Swift 2025
B_{crit}	4.4×10^{13} T	QED critical
B/B_{crit}	$\sim 10^{-3}$ to 10^{-2}	Calibrated
Mass M_{ns}	≈ 1.4 M?	Standard NS
Galactic center distance	~ 0.1 pc from Sgr A*	VLBI

UQFF equations: $g_{\text{Magnetar}}(r,t) = (GM/r^2)(1+H(z)\cdot t)(1-B/B_{crit})+(GM_{BH}/r_{BH}^2)+U_{g1}+U_{g2}+U_{g3}+U_{g4}+\Lambda c^2/3$
Verification: Chandra/Swift 2025, PAPER_013 (spin-down), PAPER_121 (full equation)
See: PAPER_121 for the complete 50+ term expansion

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10. Galactic Structural Systems

10.1 Galactic Disk

Role in UQFF: Stability analysis for $U_{b,i}$, galactic rotation curve.

Parameter	Value	Source
ω_g	7.3×10^{-16} rad/s	Galactic rotation
v_{gal}	220 km/s	Milky Way rotation curve
r_{8kpc}	8 kpc = 2.47×10^{20} m	IAU 2012

UQFF equations: $U_{b,i}$ stability, U_{g4} BH interaction
Verification: Gaia DR4 2025

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11. AGN and Cosmic Ray Systems

11.1 LLAGNs (Low-Luminosity AGNs)

Role in UQFF: CRP neutrino flux at ∼IceCube background level.

Parameter	Value	Source
Neutrino flux	∼IceCube background 10^{-18} GeV?¹cm?²s?¹sr?¹	IceCube 2022
Dominant process	pp collisions < 0.1 PeV	EP-10
SED peak	0.05 PeV	EP-10 PAPER_108
Spectral index	$\Gamma \approx 2.37$	IceCube HESE

UQFF equations: CRP term $\sum D_E \partial^2 n/\partial p^2 \cdot e^{-\gamma t}$, pp/p? SED
Verification: EP-10 (PAPER_108), IceCube 2022

11.2 High-Energy Cosmic Ray Sources

Role in UQFF: CRP p_{max} calibration.

Parameter	Value	Source
p_{max}	10^{16} eV	Fermi/Chandra 2025
$n(p) \propto$	$p^{-2.2} e^{-p/p_{max}}$	Fokker-Planck solution
$D_E \propto$	$E^{0.5}$ (Kolmogorov turbulence)	CRP physics

UQFF equations: Fokker-Planck equation, CRP term in F_U
Verification: Fermi/Chandra/Parker 2025

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12. Cosmological Systems

12.1 Solar Flares

Role in UQFF: U_{g1} dipole calibration, $B_s = 0.4$ T upper bound.

Parameter	Value	Source
B_s (flare)	0.4 T	GOES 2025
B_s (quiet)	0.0001 T	GOES baseline

UQFF equations: U_{g1} dipole term
Verification: GOES 2025

12.2 Intergalactic Medium

Role in UQFF: ρ_{vac} ratio calibration for vacuum hierarchy.

Parameter	Value	Source
ρ_{vac} ratios	$\sim 10^{-38}$ (low/high components)	EP-08
λ_{vac} range	$\sim 10^{-9}$ J/m³ (DE level)	JCAP 2025
DM density	~ 0.47 GeV/cm³ = 8.4×10^{-25} J/m³	JCAP01(2025)021
Primordial DM	$\sim 10^{-26}$ J/m³	JCAP07(2025)033

UQFF equations: $\lambda_{vac} = \sum(f_i E_i)/V$, ρ_{vac} alignment
Verification: EP-08 (PAPER_118), JCAP 2024/2025

12.3 Quasar Jets (General Class)

Role in UQFF: Fluid/Navier-Stokes jet modeling with $U_{b,i}$ asymmetry.

Parameter	Value	Source
Jet velocity	$\sim 0.7c$–$0.99c$	VLBI proper motion
Asymmetry mechanism	$\cos(\omega t_n)$ reversal	UQFF $U_{b,i}$

UQFF equations: $U_{b,i}$, Navier-Stokes in jet fluid
Verification: Chandra/MNRAS general

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13. Q_wave_47 Statistical Summary

The 47-system Q_wave distribution (from the UQFF verification corpus) encompasses a subset of the 24 systems above plus additional intermediate systems:

Statistic	Value
Mean $\bar{Q}_{wave}$	3.97×10^4 J/m³
Standard deviation	5.11×10^4 J/m³
Jarque-Bera	8.78 ($p = 0.012$)
Leptokurtosis	0.037
Distribution	Non-normal (heavy-tailed)
N systems	47 (superset of 24 catalog systems)

The non-normal distribution confirms that UQFF Q_wave energy is not uniformly distributed across system types — triadic/turbulent systems (Westerlund 2, Pillars) contribute heavy tails, while individual systems (LLAGNs, nuclear) contribute near-zero floor values.

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14. Cross-Reference Table: Systems by Whitepaper

System	Type	Primary Paper	EP
Sun	Stellar	PAPER_114	EP-07
Westerlund 2	Star cluster	Q_wave_47 corpus	—
Sgr A*	SMBH	PAPER_110	EP-06
G359.13142-0.20005	High-z	JWST corpus	—
3C 273	Quasar	PAPER_115	EP-09
RACS J0320-35	Quasar	PAPER_111	EP-01
Blazars (4LAC)	Blazars	PAPER_113	EP-05
PLCK G287.0+32.9	Galaxy cluster	Triadic corpus	—
ASKAP J1832-0911	Radio transient	Triadic corpus	—
PSZ2 G181.06+48.47	Galaxy cluster	Triadic corpus	—
GW170817	BNS merger	PAPER_109	EP-11
AT2024tvd	Transient	Triadic corpus	—
TOI 1227 b	Exoplanet	Triadic corpus	—
Pillars of Creation	Nebula	Triadic corpus	—
Nebular Dynamics	Generic nebula	Framework	—
Pb-206	Nuclear	PAPER_117	EP-04
^{12}C Hoyle State	Nuclear BEC	PAPER_107	EP-12
SGR 1745-2900	Magnetar	PAPER_013/121	—
Galactic Disk	Galactic	PAPER_110	—
LLAGNs	AGN class	PAPER_108	EP-10
CR Sources	High-energy	PAPER_088	—
Solar Flares	Sun	PAPER_114	EP-07
Intergalactic Medium	Cosmological	PAPER_118	EP-08
Quasar Jets	AGN class	PAPER_111/115	EP-01/09

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